Glycan-Reactive Innate-like B Cells and Developmental Checkpoints

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Glycan-Reactive Innate-like B Cells and Developmental Checkpoints

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Using an Ig H chain conferring specificity for N-acetyl-D-glucosamine (GlcNAc), we developed transgenic (V_HHGAC39 TG) mice to study the role of self-antigens in GlcNAc-reactive B-1 B cell development. In V_HHGAC39 TG mice, GlcNAc-reactive B-1 B cell development during ontogeny and in adult bone marrow was normal. However, adult TG mice exhibited a block at transitional-2 immature B cell stages, resulting in impaired allelic exclusion and accumulation of a B cell subset coexpressing endogenous Ig gene rearrangements. Similarly, V_HHGAC39 B cell fitness was impeded compared with non-self-reactive V_HJ558 B TG cells in competitive mixed bone marrow chimeras. Nonetheless, adult V_HHGAC39 mice immunized with *Streptococcus pyogenes* produce anti-GlcNAc Abs. Peritoneal cavity B cells transferred from V_HHGAC39 TG mice into RAG$^{-/-}$ mice also exhibited robust expansion and anti-GlcNAc Ab production. However, chronic treatment of young V_HHGAC39 mice with GlcNAc-specific mAbs leads to lower GlcNAc-binding B cell frequencies while increasing the proportion of GlcNAc-binding B1-a cells, suggesting that Ag masking or clearance of GlcNAc Ags impedes maturation of newly formed GlcNAc-reactive B cells. Finally, BCR H chain editing promotes expression of endogenous nontransgenic BCR alleles, allowing potentially self-reactive TG B cells to escape anergy or deletion at the transitional stage of precursor B cell development. Collectively, these observations indicate that GlcNAc-reactive B cell development is sensitive to the access of autologous Ags. *The Journal of Immunology*, 2024, 212: 1913–1921.

Innate-like B-1 B cells constitute an important frontline defense against infectious pathogens and contribute to host immunity by producing natural Ab in serum and mucosal secretions through their intrinsic ability to rapidly differentiate into Ab-secreting cells (ASCs) following Ag challenge. Biased Ig gene recombination and restricted junctional diversity drive B-1 B cell generation through a unique developmental program during fetal and neonatal life. This program in concert with key selection checkpoints gives rise to canonical, and often self-reactive, B cell receptor clonotypes, although neither is required for B-1 B cell development or spontaneous natural IgM secretion (1). Microbial Ags play a poorly understood role in shaping B-1 B cell repertoire development and natural antibody specificities in the preimmune Ig repertoire (2), but they are nonetheless important drivers of microbe-reactive B cell development.

In both mice and humans, many natural Abs (NAbs) are directed against carbohydrate Ags, and serum Abs reactive with N-acetyl-D-glucosamine (GlcNAc) epitopes comprise a significant portion of the NAb repertoires (3). GlcNAc is a highly conserved antigenic epitope found on multiple bacterial, fungal, and protozoan pathogens (4); it is the immunodominant epitope of the Lancefield group A carbohydrate (GAC) cell wall Ag expressed by *Streptococcus pyogenes* (group A *Streptococcus* [GAS]) (5).

We and others have shown that anti-GlcNAc Abs elicited by immunizing mice with GAS recognize self-antigens generated through post-translational glycosylation, including cytoplasmic O-GlcNAc modifications and cryptic N-linked glycan epitopes (2, 6, 7). Like mice, most humans possess relatively high titers of serum GlcNAc-specific Abs in the absence of known pathologies. GlcNAc-reactive Abs reactive with neurons have been implicated in neurologic sequelae in some patients following GAS infection (8), demonstrating that certain GlcNAc-reactive B cell clonotypes reactive to bacterial Ags might be pathogenic due to their specificity for self-antigens.

We recently demonstrated that in mice, microbiota-derived Ags may be essential drivers of Ag-specific B cell development. GlcNAc-reactive B cells, which are absent in germ-free mice, populate the B-1 B cell compartment and dominate immune responses to GAS (9). In contrast to the microbiota-independent development of canonical B-1 B cells reactive with self-antigens such as phosphorylcholine (10), intestinal microbiota endogenous GlcNAc epitopes promote the establishment and expansion of GlcNAc-reactive B-1 B cell clonotypes (9). Because GlcNAc is an auto-antigen, we sought to evaluate the effects of self-antigens on GlcNAc-reactive B cell development and expansion. We developed a IgH transgenic (TG) mouse expressing a germline V(D)J rearrangement derived from the GlcNAc-specific HGAC39 hybridoma, which is reactive to both microbial glycans and autoantigens expressing GlcNAc epitopes. Here, we show that expression of a GlcNAc-specific BCR disadvantages B cells at peripheral tolerance checkpoints, revealing an important restriction in the development of B cells specific for autologous carbohydrate Ags.

Materials and Methods

Animals

V_HHGAC39 TG mice were generated through the University of Alabama at Birmingham Transgenic & Genetically Engineered Models Core Facility using previously detailed methodologies (11). Briefly, the V_H6-6-D_H1-1J_H3 VDJ rearrangement was amplified from genomic DNA of the HGAC39 IgG3 hybridoma (12) with primers specific for the putative signal peptide (5′-CGCCATGGACTTGAGACTGAGCTG-3′) and the JH3 intron

Department of Microbiology, University of Alabama at Birmingham, Birmingham, AL

ORCIDs: 0000-0002-2034-2353 (B.L.P.D.); 0000-0003-2569-2560 (J.F.K.); 0000-0001-6522-0053 (R.G.K.).

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John F. Kearney is a Distinguished Fellow of AAI.

Address correspondence and reprint requests to Dr. John F. Kearney, University of Alabama at Birmingham, 1825 University Boulevard, Shelby Biomedical Research Building, Room 410, Birmingham, AL 35294-2812. E-mail address: jfk@uab.edu

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Abbreviations used in this article: ASC, Ab-secreting cell; BM, bone marrow; GAC, group A carbohydrate; GAS, group A Streptococcus; GlcNAc, N-acetyl-d-glucosamine; LM, littermate; MZ, marginal zone; Nab, natural Ab; p.i., postinfection; TG, transgenic; WT, wild type.

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(5′-GACGGATCCCCATGTCTCAACTT TGGG-3′). This PCR product was ligated into the KC-7 vector (10), following digestion at Nco1 and BamH1 restriction sites. The resulting vector sequence was verified by nucleotide sequencing, and an L chain–producing B cell line was used to confirm successful in vitro expression of the KC-7-HGAC39 construct. Last, plasmid DNA was removed via digestion at SalI sites flanking the transgene, and the resulting product was microinjected into C57BL/6/DBA F2 fertilized eggs. Founder mice were identified by PCR screening for transgene integration, resulting in three founder strains. One strain was selected and further backcrossed onto C57BL/6 mice for >20 generations. V_HJ558 and V_H81x Tg mice have been described previously (13, 14). All V_H TG mice weremaintained at hemizygosity, and non-TG littermates (LMs) were used as experimental controls. $Rag1^{-/-}$ (B6.129S7-$Rag1^{tm1Mom}$/J) mice and CD45.1 C57BL/6 (B6.SJL-Ptprca Pepcb/BoyJ) mice were purchased from The Jackson Laboratory. All mice were provided food and acidified water ad libitum and were bred and housed in specific pathogen-free facilities according to rules and regulations established by the University of Alabama Animal Resource program. All mice were used at 8–12 wk of age except those described in Fig. 4B, where mAb administration was begun at 17 d of age. For GAS immunization, mice 8–12 wk of age were injected i.p. with 1e8 CFU equivalents of heat-killed, pepsin-treated GAS vaccines prepared as described by New et al. (11) using GAS strain J17A (American Type Culture Collection, 12385).

Hybridoma maintenance

Igh6 (IgM^b) allotype-specific mAb was derived from previously described MB86 (15) and RS-3.1 (16) mouse IgG1 anti-mouse IgM^a hybridomas. GlcNAc-reactive Ig-restricted idiotype-specific mAbs were derived from the IA-1 and 4H-1 hybridomas (17), generously provided by Dr. Neil Greenspan (Case Western Reserve University). All IgG hybridomas were grown in Hybridoma SF media (Life Technologies) and mAbs were purified from culture supernatants with protein G-Sepharose. Anti-GlcNAc IgM HGAC78 (12) and anti-hen egg lysozyme MD4 were isolated from RS-3.1-Sepharose columns. mAbs were concentrated using Centricon Plus-70 centrifugal filters (Millipore). mAb preparations were quantitated by spectrophotometric analysis (ND-1000, NanoDrop Technologies). Quality was validated by SDS-PAGE and Ag-binding assays plus limulus assay (Amebocyte Lysate Pyogent, Lonza). In some cases, mAbs were directly conjugated to Alexa Fluor fluorescence reagents (Life Technologies).

Flow cytometry

For analyzing B cell surface phenotypes and Ig allotype expression, 5×10^6–10^7 cells were pelleted in 96-well round-bottomed microtiter plates (Costar) and blocked with anti-CD16/CD32 (Ab93) prior to staining with B cell marker–specific mAbs obtained from BD Biosciences, including anti-CD19 (1D3), B220 (RA3-6B2), IgM (R6-60.2), IgD (11-26c.2a), CD93 (AA4.1), CD24 (M1/69), Ly-51 (BP-1), CD43 (S7), CD5 (53-7.3), CD23 (B3B4), CD21 (7G6), CD1d (1B1), CD11b (M1/70), and CD138 (281-2). mAbs IA-1 and 4H-1 were used to detect Id3a and Id3b idiotopes, respectively (16). Fluorescent polysaccharides, including GAC peptidoglycan (PG-PS 10s, Lee Labs) and α(1, 3)-dextran (DEX; a gift from Dr. A. Jeanes) were used to detect Ag-reactive B cells through single-epitope, dual-color staining as described previously (18). Flow cytometry was performed on an LSRII or a FACSAria (Custom Configuration) at the University of Alabama at Birmingham Comprehensive Flow Cytometry Center.

Adoptive transfer

To generate mixed bone marrow (BM) chimeric mice, we used previously established parameters. Briefly, recipient mice were lethally irradiated with two doses of 750 Gy using an X-ray irradiator. Animals were subsequently administered BM mixtures composed of 50% CD45.1+ BM and 50% CD45.2+ VH transgene expressing BM. Animals were subsequently housed for 8 wk to allow BM reconstitution prior to analysis.

Statistical analysis

For each experiment, means and SEMs were determined for the measured parameter, and single-factorial (Student *t* test) or multifactorial (one- or two-way ANOVA) statistical comparisons were made. Sample sizes for mouse experiments were based on preliminary data and previous reports such that appropriate statistical testing could yield sufficient power. In all cases, similar variance for sample groups was ensured, and assumptions of statistical comparisons were met. No specific randomization or exclusion criteria were applied due to use of inbred mouse strains, and blinding was not performed during allocation of mice to experimental groups. All analyses were completed using GraphPad Prism software (GraphPad Software).

Results

V_HHGAC39 mice develop a restricted B cell repertoire enriched for GAC-reactive B-1 B cells

We generated V_HHGAC39 C57BL/6 TG mice by cloning the rearranged IGH gene of the GlcNAc-reactive HGAC39 hybridoma (γ3) (12) into the KC-7 expression vector, which drives BCR transgene expression in developing B cells via the V_H81x promoter (11). Greater than 80% of circulating B cells expressed transgene-encoded $Igh6a^+$ (IgM^a) BCR, and allelic exclusion was intact except for small populations of B cells staining for both IgM^a and the endogenous Igh6b (IgM^b) BCR that were highest in splenic and PerC B cells (Fig. 1A, 1B, Supplemental Fig. 1A).

The (PerC) B cells in V_HHGAC39 mice had decreased GAC^+ $CD5^+CD23^-$ B-1a B cells and higher frequencies of B1b B cells (Supplemental Fig. 1B). B-1 B cell populations in V_HHGAC39 PerC were enriched for B cells expressing endogenous IgM^b gene rearrangements or coexpressing IgM^b and transgene-encoded BCR (Supplemental Fig. 1C). V_HHGAC39$^{+/0}$ mice exhibited a disproportionate expansion of splenic $CD21^{Hi}CD23^{Lo}$ marginal zone (MZ) B cells (Supplemental Fig. 1B). By contrast, splenic B cells expressing endogenous gene rearrangements (identified through the expression of IgM^b) were predominantly localized to the B cell follicles (Fig. 1C).

Despite >30-fold higher frequencies of GAC-binding B cells in the spleen and PerC of V_HHGAC39 compared with non-Tg LM mice (Fig. 1D), V_HHGAC39$^{+/0}$ mice did not exhibit higher levels of resting serum anti-GAC Abs (Fig. 1E). Compared with non-TG mice, GAC-binding B cells were disproportionately $CD23^{high}$ in V_HHGAC39 mice (Fig. 1F) and uniformly expressed the HGAC39 mAb idiotypes IdI-3a and IdI-3b (Fig. 1G) (18). Despite their restricted idiotype profile, V_HHGAC39$^{+/0}$ GAC^+ B cells were phenotypically heterogeneous, but, unlike non-TG GAC-biding B cells, they did not express CD43 (Fig. 1H).

Overall, in the VHGAC39 mice, GAC+ B cells were distributed in the MZ, FO, and B-1 B cell compartments relative to GAC+ B cell distribution in non-TG LM mice. We therefore concluded that the VHHGAC39 transgene drove establishment of a clonally restricted population of GAC-binding B cells that localized to innate-like B cell compartments but exhibited surface phenotypes dissimilar to those of wild-type (WT) mice and canonical B-1 B cell phenotypes.

V_HHGAC39-expressing B cells respond to Streptococcus pyogenes *immunization*

The absence of increased resting anti-GAC serum Abs despite elevated numbers of anti-GAC B cells along with an absence of CD43 expression by V_HHGAC39$^{+/0}$ GAC^+ B cells led us to investigate the functionality of TG B cells. Following i.p. immunization of V_HHGAC39$^{+/0}$ mice with a GAS vaccine, IgM^a TG B cells visualized with RS-3.1 translocated through bridging channels to splenic red pulp areas. In contrast, immunization of V_HHGAC39$^{+/0}$ mice with *Streptococcus dysgalactiae* (group C *Streptococcus*), which synthesizes N-acetyl-D-galactosamine–bearing cell wall carbohydrate, did not elicit a similar B cell response (Fig. 2A), indicating Ag-specific TG B cell activation. Levels of IgM anti-GlcNAc and anti-GlcNAcBSA Abs in V_HHGAC39$^{+/0}$ mice immunized with the GAS vaccine, although high, with endpoint titers in excess of 1/6400, were slightly reduced compared with immunized non-TG LMs (Fig. 2B). The number of GAC-binding B cells increased significantly following GAS immunization in the spleen and PerC of V_HHGAC39$^{+/0}$, suggesting Ag-specific B cell proliferation (Fig. 2C). This observed expansion corresponded with the emergence of a large number of GAC-binding $CD138^+$ cells indicative of

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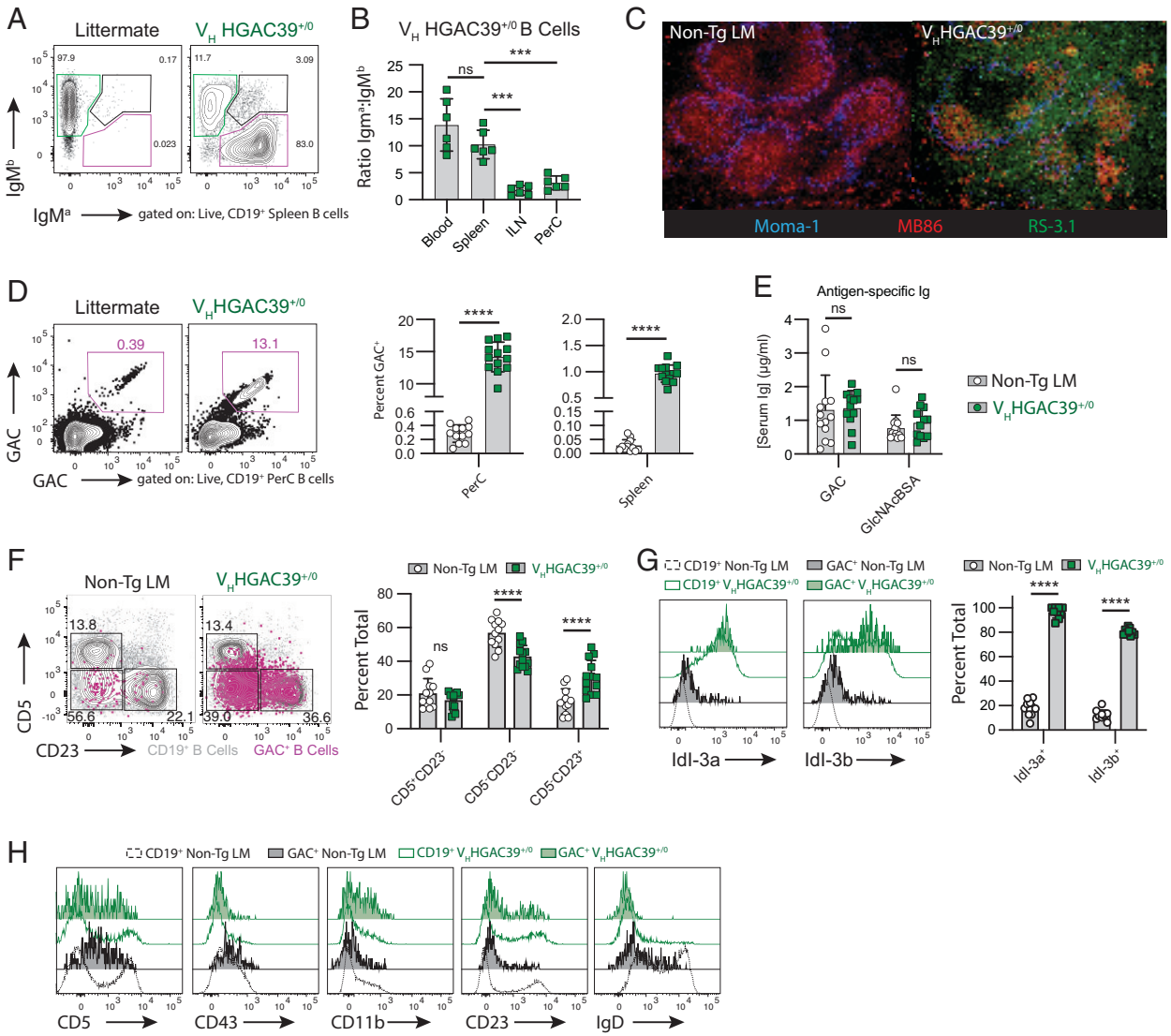


FIGURE 1. The V_H HGAC39 BCR transgene drives production of clonally restricted GAC-reactive B cells. (**A**) Representative flow cytometric analysis of IgM allotype expression in splenic B cells of age-matched non-TG LM and $V_HHGAC39^{+/0}$ mice, using mAbs MB86 (IgM^b), RS-3.1 (IgM^a) and Moma-1. (**B**) Quantification of the ratios of transgene-encoded (IgM^a) and endogenous (IgM^b) BCR in the indicated tissues of $V_HHGAC39^{+/0}$ mice ($n = 6$). Data are mean ± SD of two independent experiments with individual data points overlaid from one experiment. ***$p < 0.001$ by one-way ANOVA with Tukey's multiple comparisons test. (**C**) Representative immunofluorescence images (original magnification ×40) of splenic lymphoid follicles in non-Tg LM and $V_HHGAC39^{+/0}$ mice. The distribution of IgM^aV_H HGAC39-expressing B cells (green), endogenous B cells IgM^b (red), and MZ macrophages (Moma-1) (blue) are shown. (**D**) Representative flow cytometric analysis of GAC staining Perc B cells in non-TG LM control and $V_HHGAC39^{+/0}$ mice (left) and quantitation of GAC staining B cells in Perc and spleen 10–12 wk of age. ****$p < 0.0001$, by two-way ANOVA (right). (**E**) Ag-specific Ab concentrations in non-Tg LM ($n = 13$) and $V_HHGAC39^{+/0}$ mice ($n = 12$). Two-way ANOVA with Šídák's multiple comparisons test. (**F**) Flow cytometric analysis of phenotypes of CD19 and GAC+ Perc B cells from non-TG LM control mice and $V_HHGAC39^{+/0}$ (left) and B cell subset enumeration (right). ****$p < 0.0001$ by two-way ANOVA. (**G**) Representative flow cytometric analysis of splenic GAC-binding B cells of $V_HHGAC39^{+/0}$ mice with the indicated anti-idiotype mAbs (left) and quantification of idiotype$^+$ populations in 10–12-wk-old $V_HHGAC39^{+/0}$ and non-TG LM control mice (right); data are mean ± SEM from one representative experiment completed twice; ***$p < 0.001$ by one-way ANOVA. (**H**) Flow cytometric analysis of phenotypes of CD19 plus GAC+ Perc B cells from $V_HHGAC39^{+/0}$ and non-TG LM control mice. ****$p < 0.0001$ by two-way ANOVA.

differentiation into plasma cells following immunization with GAS (Fig. 2D). The slightly reduced levels of GAC-reactive Ab present in V_H HGAC39$^{+/0}$ mice compared with LM despite the expansion of plasma cells led us to evaluate the Ab secretion of TG B cells. Following immunization, splenic B cells from $V_HHGAC39^{+/0}$ and LM control animals were subjected to ELISPOT analysis to determine the number of B cells secreting GAC-reactive Ab. Indeed, the numbers of GAC-binding plasma cells were significantly elevated in TG mice, although the spot size appeared much smaller in samples derived from $V_HHGAC39^{+/0}$, indicative of a reduced rate of Ab production in TG B cells (Fig. 2E). Collectively, these results demonstrate that Ag receptor signaling was intact in $V_HHGAC39^{+/0}$ B cells that responded equivalently to WT B cells despite their increased frequency, although the expression of the IgM TG appeared to reduce the Ab production by Ag-reactive plasma cells.

Because the Ag-reactive B cell frequency is significantly elevated in $V_HHGAC39^{+/0}$ mice and we previously showed that B-1 B cells are the major precursors for GlcNAc-reactive ASCs induced by GAS immunization (9), we next investigated whether transferred PerC B cells would convey responsiveness to an immunodeficient host. In this experiment, CD19$^+$ PerC B cells were magnetically enriched on the basis of CD19 expression from either

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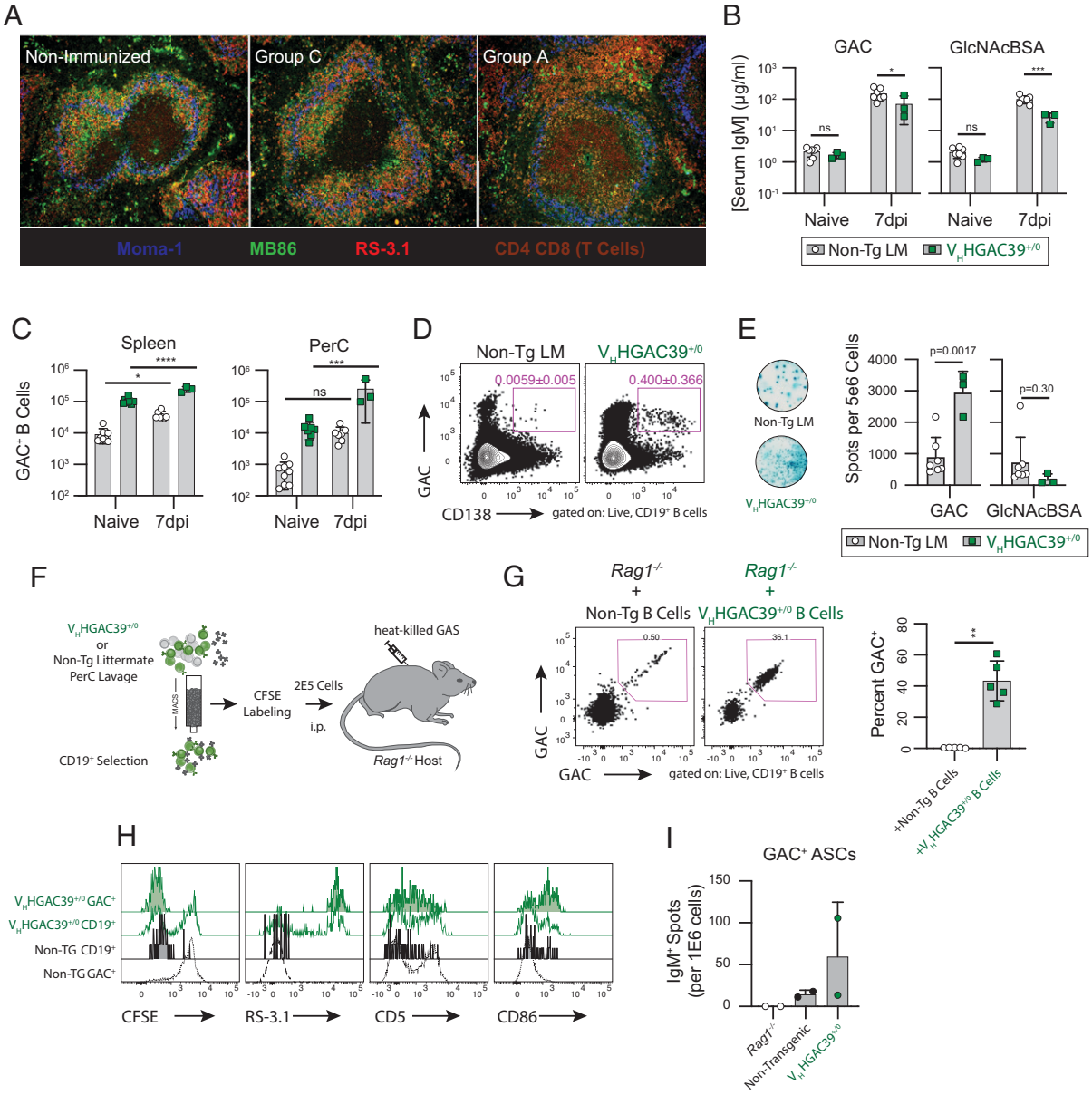


FIGURE 2. $V_HHGAC39^{+/0}$ B cells are responsive to GAS immunization. (**A**) Immunofluorescence imaging (original magnification ×40) of splenic lymphoid follicles from $V_HHGAC39^{+/0}$ mice either nonimmunized (left) or immunized with group C (center) or group A streptococci (right) stained with mabs Moma-1, MB86, RS-3.1, H129.19 (anti-mCD4), and 53-6.6 (anti-mCD8). (**B**) Serum concentration of GAC- and GlcNAcBSA-specific Ab in non-TG LM mice (white, $n = 7$) and $V_HHGAC39^{+/0}$ mice (green, $n = 3$) before and 7 d postimmunization (d PI) with GAS. (**C**) Enumeration of GAC- and GlcNAcBSA-specific ASCs in non-TG LM (white, $n = 7$) and $V_HHGAC39^{+/0}$ (green, $n = 3$) mice 7 d postimmunization (d PI) with GAS. (**D**) Representative flow cytometric histograms of GAC-binding $CD138^+$ plasma cells in spleens of either $V_HHGAC39^{+/0}$ or LM mice 7 d PI with GAS. (**E**) Increases in GAC-binding B cell numbers in spleen and PerC of non-Tg LM (white) and $V_HHGAC39^{+/0}$ (green) mice 7 d. PI with GAS. (**F**) Schematic depicting the B cell enrichment strategy of non-TG LM and $V_HHGAC39^{+/0}$ PerC B cells, CFSE labeling, and adoptive transfer to lymphocyte-deficient *Rag1*$^{-/-}$ recipients. (**G**) (left) Representative flow cytometric analysis of PerC-localized GAC^+ donor B cells in *Rag1*$^{-/-}$ recipient mice transferred non-TG LM or V_H $HGAC39^{+/0}$ enriched PerC B cells (left) and enumeration of GAC^+ donor B cell frequency (right) ($n = 5$). (**H**) Proliferation measured by CFSE dilution of the indicated population of donor B cells 7 d following immunization with GAS. (**I**) Number of donor B cell–derived GAC-reactive ASCs determined by ELISPOT.

$V_HHGAC39^{+/0}$ or non-TG LM mice, CFSE labeled, and adoptively transferred into *Rag1*$^{-/-}$ mice, which were then immunized i.p. with GAS (Fig. 2F). Flow cytometric analysis of PerC cells in recipient mice 3 d postinfection (p.i.) revealed large numbers of GAC binding B cells in the V_H $HGAC39^{+/0}$ transferred mice (Fig. 2G). Analysis of CFSE labeling revealed extensive proliferation of GAC-binding B cells derived from both non-TG LM and $V_HHGAC39^{+/0}$ B cells (Fig. 2H). TG B cell expansion again correlated with an increase in the numbers of GAC-reactive ASCs (Fig. 2I). Collectively, these results confirm V_H $HGAC39^{+/0}$ B cells are Ag responsive in vivo because GAC-binding B cells are mobilized following immunization, upregulate activation markers, and differentiate in ASCs.

Peripheral tolerance checkpoints drive negative selection of $V_HHGAC39$-expressing B cells in adult mice

Although GAC-reactive $V_HHGAC39^{+/0}$ B cells were functionally responsive to antigenic stimulation, reduced B cell numbers and incomplete allelic exclusion observed in $V_HHGAC39^{+/0}$ mice

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suggested that expression of the $V_HHGAC39$ transgene disrupts B cell development. Indeed, we observed a significant decline in IgM^a-expressing peripheral blood B cells with a concordant increase in IgD^+ B cells with age in $V_HHGAC39^{+/0}$ mice, indicative of the accumulation of B cells expressing endogenous IG rearrangements (Fig. 3A). Moreover, the frequency of endogenous IGH gene rearrangement observed in adult $V_HHGAC39^{+/0}$ B cells exceeded that observed in $V_HJ558^{+/0}$ and $V_H81x^{+/0}$ mice harboring similar IGH-KC7 IG gene TG constructs (Fig. 3B), implicating the specificity of $V_HHGAC39$ BCR (11, 13) in this phenomenon. Because the V_HJ558 TG line expresses an IGH transgene derived from the DEX-specific J558 plasmacytoma that is not known to react with an auto-antigen (13), we next compared developmental checkpoints in immature B cells isolated from $V_HHGAC39^{+/0}$ and $V_HJ558^{+/0}$ mice.

As expected in TG mice expressing a rearranged IGH, both $V_HHGAC39^{+/0}$ and $V_HJ558^{+/0}$ mice exhibited modestly increased frequencies of Pre-Pro B cells (Fr. A) relative to non-TG LM mice. In both strains, a majority of Fr. E ($B220^+BP\text{-}1^-CD24^+IgM^+$) immature B cells expressed the TG IgM^a BCR transgene (Supplemental Fig. 2A, 2B). Although both TG strains exhibited an overall decrease in the frequency of recirculating Fr. F ($B220^{Hi}BP\text{-}1^-CD24^+IgM^+$) B cells in BM, $V_HHGAC39^{+/0}$ mice displayed a higher frequency of mature B cells expressing the endogenous IgM^b compared with $V_HJ558^{+/0}$ mice (Supplemental Fig. 2C), whereas the frequencies of Fr. E and immature B cells were similar in both of these TG strains. These data suggested that late B cell developmental checkpoints may be differentially affected by these TGs. Flow cytometric histograms of $CD93^+$ $B220^+$ immature splenic B cells, and IgM^+CD23^- transitional B cell subpopulations in $V_HJ558^{+/0}$ and $V_HHGAC39^{+/0}$ (Fig. 3C) revealed a significant reduction in Tr.2 population frequencies in $V_HHGAC39^{+/0}$ compared with $V_HJ558^{+/0}$ splenic B cells, with an increase in Tr.3 B cell frequencies (Fig. 3D). Additionally, Tr.2 B cells derived from $V_HHGAC39^{+/0}$ mice exhibited a 4-fold increase in the percentage of cells expressing endogenous IgM^b BCR (Fig. 3E and Supplemental Fig. 2D). These findings suggest that peripheral tolerance checkpoints impair the progression of $V_HHGAC39^{+/0}$ transitional B cell populations to maturity due to self-reactivity of endogenous GlcNAc epitopes.

V_H HGAC39-expressing B cells are positively selected into the B-1 B cell compartment during early life

In contrast to T1 B cells, which are relatively unresponsive to BCR stimulation, T2 B cells upregulate survival factors following BCR ligation. Accordingly, the loss of BCR transgene expression observed in $V_HHGAC39^{+/0}$ T2 B cells could be explained by BCR signaling derived from GlcNAc-modified self-antigens, which we have shown are relevant in the NOD autoimmune mouse (7), encountered during peripheral maturation. To determine the feasibility of this hypothesis, we evaluated the accessibility of endogenous GlcNAc Ags by monitoring the clearance of passively administered Ab. Purified GlcNAc-

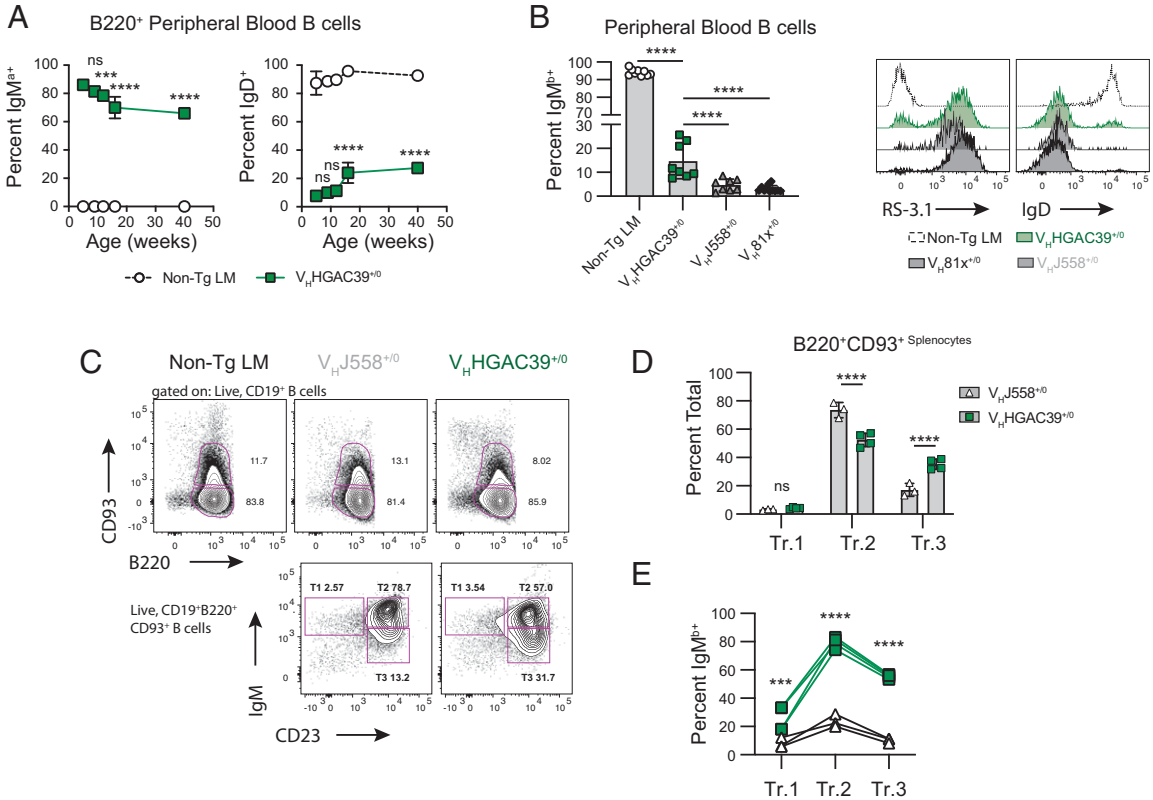


FIGURE 3. Peripheral tolerance checkpoints impede $V_HHGAC39$-expressing B cell development. (**A**) (left) Decline in IgM^a (right) and increased IgD expression peripheral blood B cells with age in $V_HHGAC39^{+/0}$ mice. (**B**) (left) Percentages of peripheral blood B cells that express endogenous IgM^b gene rearrangements in the indicated mouse strains. Data are plotted as mean ± SD with individual data points overlaid, pooled from multiple experiments. $^{****}p < 0.0001$ by one-way ANOVA with Tukey's multiple comparisons test. (right) Representative flow cytometric analysis of IgM^a transgene and IgD expression on $B220^+$ peripheral blood cells of mice from the indicated strain. (**C**) Representative flow cytometric histograms of $CD93^+B220^+$ immature splenic B cells, and IgM^+CD23^- transitional B cell subpopulations in $V_HJ558^{+/0}$ and $V_HHGAC39^{+/0}$ mice. (**D**) Relative frequencies of splenic T1, T2, and T3 immature B cell subpopulations in $V_HJ558^{+/0}$ ($n = 3$) and $V_HHGAC39^{+/0}$ ($n = 4$) mice. (**E**) Percentages of peripheral blood B cells expressing endogenous IgM^b gene rearrangements in the indicated splenic transitional immature B cell subpopulations of $V_HJ558^{+/0}$ ($n = 3$) and $V_HHGAC39^{+/0}$ ($n = 4$) mice. Data are mean ± SD with individual data points overlaid from one representative experiment completed twice. $^{****}p < 0.0001$ by one-way ANOVA with Tukey's multiple comparisons test.

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specific (HGAC78) or hen egg lysozyme–specific (MD4) IgM mAbs were passively administered to WT C57BL/6. The clearance of each mAb in the individual mice was determined by evaluating serum IgM^a by ELISA. HGAC78 rapidly disappeared from WT C57BL/6 serum (12) compared with MD4, which does not exhibit significant self-reactivity (Fig. 4A). The observed difference in clearance of these IgM Abs suggests that GlcNAc epitopes are bound by passive IgM Ab and, by inference, B cells expressing GlcNAc-specific BCRs (Fig. 4C). Collectively, these results suggest that $V_HHGAC39^{+/0}$ TG BCRs are unfavorable for B cell progression through transitional 2 B cell peripheral tolerance checkpoints, promoting secondary endogenous IGH gene rearrangement and resulting in the coexpression of BCRs or loss the $V_HHGAC39$ H chain.

To evaluate the impact of GlcNAc-bearing auto-antigens on $V_HHGAC39$ B cell development, we treated $V_HHGAC39^{+/0}$ and non-TG LM pups with purified GlcNAc-specific IgM mAb HGAC78 or a control IgM every 3 d from days 17 to 31 of life (12) (Fig. 4B). Treatment with HGAC78 did not alter B cell subset distribution in the spleen or PerC compared with mice treated with control IgM (data not shown); however, it significantly reduced GAC-binding B cell numbers relative to control IgM-treated mice (Fig. 4C). Additionally, PerC-localized GAC^+ B cells, which were also reduced in frequency in HGAC78-treated $V_HHGAC39^{+/0}$ mice, were skewed toward a $CD5^+$ B-1a-like state compared with those treated with isotype control Ab (Fig. 4D, 4E). Interestingly, although the frequency of GAC-binding B cells was slightly reduced in non-TG control mice treated with HGAC78, the shifts in PerC B cell phenotypes were not observed. These results together show that passive GAC78 IgM Ab decreased the overall frequency and number of GAC-binding B cells but, in doing so, increased the proportion of B1a B cells. These results reflect the timing of HGAC78 treatment, which was begun 17 d after birth when TG B1a cells were preseeded into the repertoire (19). The differential sensitivity of B1b and follicular B cells suggest that HGAC78 Ab treatment impedes the development of GAC-binding cells derived from BM during the course of passive immunization.

Competition from a polyclonal repertoire restricts GAC^+ HGAC39-expressing B cells to B-1 B cell differentiation trajectories

Given the data above, we reasoned that if self-reactivity of the $V_HHGAC39$ VH transgene resulted in a developmental bottleneck, then $V_HHGAC39^{+/0}$ B cells would be competitively disadvantaged during the emergence of a polyclonal B cell repertoire. To directly test this possibility, we examined the development of B cells in mixed BM chimeras in which 50% $CD45.2^+$ $V_HHGAC39^{+/0}$ or 50% $CD45.2^+$ $V_HJ558^{+/0}$ were mixed with 50% $CD45.1^+IgM^a$ donor BM (Supplemental Fig. 3A) and transferred into irradiated CD45.1 host. Eight weeks post-transplantation, the frequency of CD45.1 and CD45.2 monocytes was determined to be ~50%, confirming effective BM chimerism (Supplemental Fig. 3B). Neither V_H $HGAC39^{+/0}$ nor V_H $J558^{+/0}$ BM recipients exhibited differences in the distribution of developing B cell subsets in BM relative to control mice reconstituted with CD45.1 BM cells (Fig. 5A, Supplemental Fig. 3C). However, the relative frequencies of CD45.2-expressing B cells represented in surface IgM^+ (immature and mature) B cells, compared with surface IgM^- B cells (up to pre-B

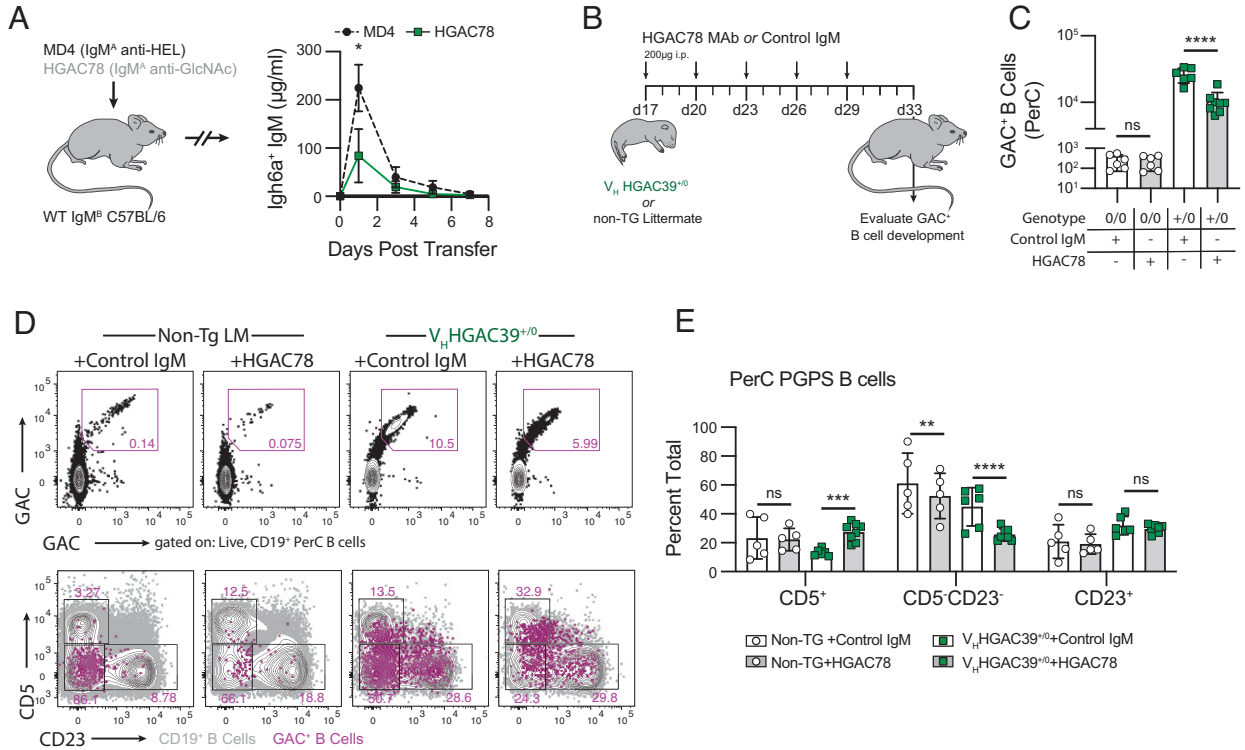


FIGURE 4. GlcNAc-bearing Ags are necessary for positive selection of GAC-reactive $V_HHGAC39^{+/0}$ B-1 B cells. (**A**) (left) Schematic depicting the passive administration of allotype-marked HGAC78 and MD4 mAbs to allogenic C57BL/6 mice, and (right) longitudinal persistence of passively administered mAbs in MD4-treated ($n = 4$) and HGAC78-treated ($n = 4$) mice as measured by ELISA. (**B**) Schematic depicting the treatment regimen of passively administered allotype-marked HGAC78 and MD4 control IgM mAbs to allogenic C57BL/6 mice. (**C**) PerC-localized GAC-binding B cell numbers in non-TG LM control mice or $V_HHGAC39^{+/0}$ mice treated with control IgM or HGAC78 during weaning. (**D**) Flow cytometric histograms of PerC GAC-binding B cells (top) and CD5 and CD23 expression of the indicated B cell populations (bottom). Data are for mice representative of each treatment group. (**E**) Relative representation of $CD5^+CD23^-$, $CD5^-CD23^-$ and $CD5^-CD23^+$ B cells in the GAC-reactive B cell population of each treatment group. Data are mean pooled from two independent experiments, with data points inlaid. Data are mean ± SD with individual data points overlaid, pooled from two replicate experiments completed twice.

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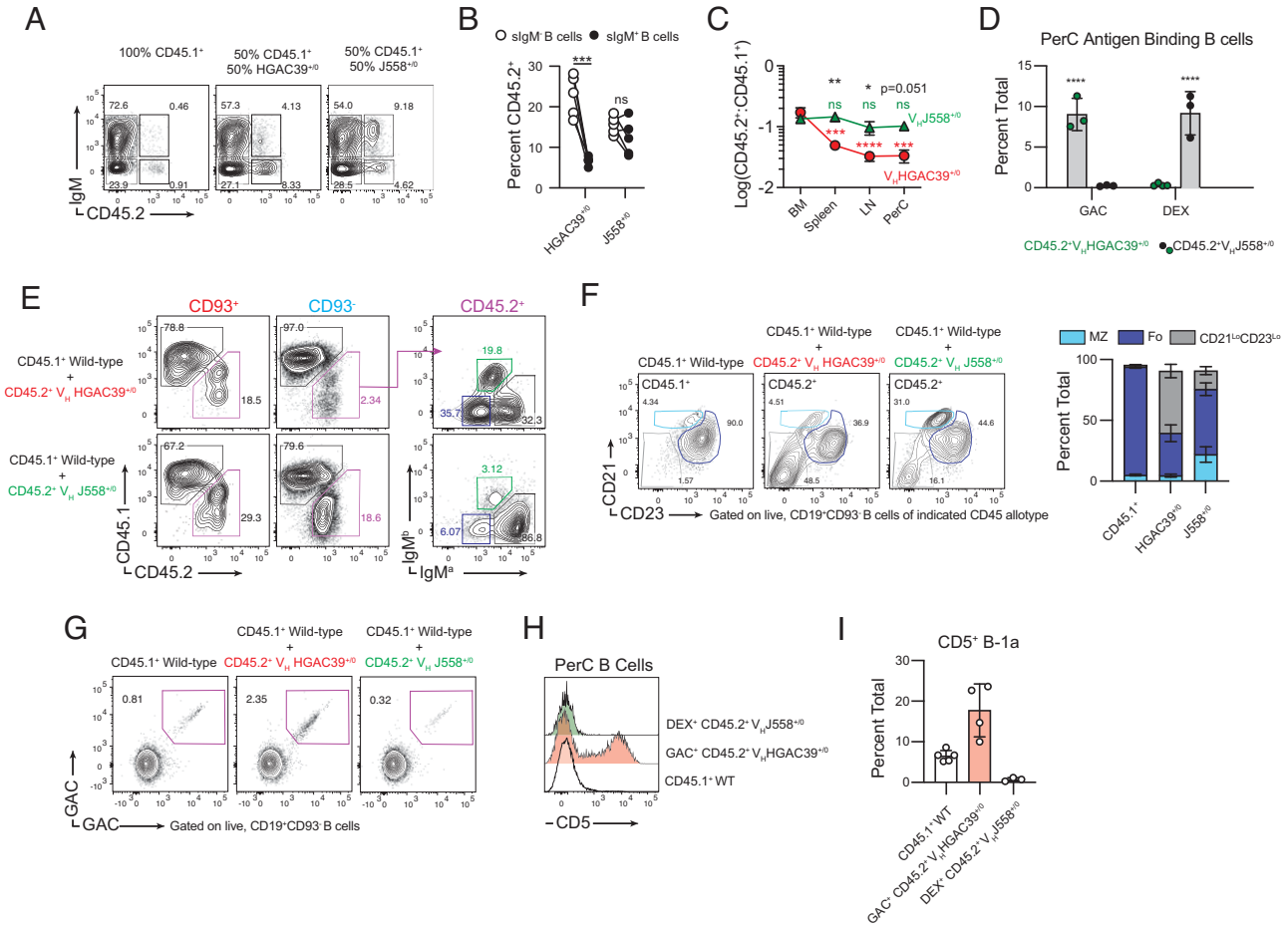


FIGURE 5. Competition from a polyclonal B cell repertoire restricts $V_HHGAC39$-expressing B cells to B-1 B cell differentiation fates. (**A**) Representative flow cytometric histograms of CD19$^+$ B cells isolated from the BM of irradiated mice 8 wk following reconstitution with BM cells from the indicated donor mice. (**B**) Frequency of CD45.2-expressing B cells in the surface IgM$^-$ (Pre-B) and surface IgM$^+$ (Fr. E + F) B cell populations of mice reconstituted with WT and $V_HHGAC39^{+/0}$ or $V_HJ558^{+/0}$ BM. Data are paired values for individual mice from one representative experiment completed twice. ***$p < 0.001$ by two-way ANOVA with Bonferroni multiple comparisons. (**C**) Ratio of CD45.2:CD45.1 B cells in the BM, spleen, lymph node, and PerC of mixed BM chimeric mice reconstituted with 50% $V_HHGAC39^{+/0}$ or $V_HJ558^{+/0}$ mouse BM. Data are mean ± SEM pooled from one representative experiment completed twice; *$p < 0.05$, **$p < 0.01$. ***$p < 0.001$, ****$p < 0.0001$ by two-way ANOVA. (**D**) Percentage of PerC B cells binding the indicated Ag in $V_HHGAC39^{+/0}$ and $V_HJ558^{+/0}$ reconstituted mice. (**E**) Representative flow cytometric analysis depicting CD45.1 and CD45.2 allotype expression in CD93$^+$ immature B cells and CD93$^-$ mature B cells (left), IgM allotype expression in CD45.2 V_H TG B cells in $V_HHGAC39^{+/0}$ and $V_HJ558^{+/0}$ mixed BM chimeric mice (right). (**F**) Representative flow cytometric histograms (left) of CD21 and CD23 expression on mature B cells and frequencies (right) of MZ, Fo, and CD21^{Low}CD23^{Low} B cells in spleen of BM chimeric mice reconstituted with WT CD45.1$^+$ BM, or 50% WT CD45.1$^+$ with either 50%CD45.2$^+$IgM^a $V_HHGAC39^{+/0}$ BM, or 50%CD45.2$^+$IgM^a $V_HJ558^{+/0}$ BM. (**G**) Flow cytometric histograms of GAC staining of B cells in PerC of BM chimeric mice reconstituted with 100% CD45.1$^+$ BM, or 50% CD45.1$^+$ admixed with either 50% CD45.2$^+$IgM^a $V_HHGAC39^{+/0}$ BM or 50% CD45.2$^+$IgM^a $V_HJ558^{+/0}$ BM. (**H** and **I**) Flow cytometric histograms (H) and frequencies (I) of CD5 expression on donor-derived CD45.1 WT B cells, GAC-binding CD45.2 $V_HHGAC39$ and DEX-binding CD45.2 V_HJ558 B cells. Data are mean ± SEM with individual data points overlaid, pooled from one representative experiment completed twice.

cell stage) were 4-fold lower in $V_HHGAC39^{+/0}$ compared with $V_HJ558^{+/0}$ BM recipients (Fig. 5B). In both V_H $HGAC39^{+/0}$ and V_H $J558^{+/0}$ BM recipients, CD19$^+$ B cells were disproportionally WT CD45.1$^+$ B cells. V_H $HGAC39^{+/0}$ expressing B cells, however, were present at 3-fold lower frequency than V_H $J558^{+/0}$-expressing B cells. Peripheral B cells isolated from PerC and lymph nodes revealed further reductions in the ratio of CD45.2$^+$:CD45.1$^+$ B cells in V_H $HGAC39^{+/0}$ BM recipients, in contrast to $V_HJ558^{+/0}$ BM recipients in which similar CD45.1:CD45.2 ratios were observed across all tissues examined (Fig. 5C). These data demonstrate that $V_HHGAC39^{+/0}$ B cells are disadvantaged during late stages of B cell development following BM egress as immature cells. However, in each case, the CD45.2 V_H $HGAC39^{+/0}$ and V_H $J558^{+/0}$ BM recipients exhibited enriched GAC and DEX Ag binding B cells in the PerC and spleen (Fig. 5D, Supplemental Fig. 3D–3F) confirming the acquisition of Ag specificity through TG expression. CD45.2-expressing cells were equivalently represented within the CD93$^+$ immature splenic B cell compartment of $V_HHGAC39^{+/0}$ and $V_HJ558^{+/0}$ BM recipient mice. However, CD45.2$^+$ cells were significantly reduced among the CD93$^-$ mature splenic B cells in $V_HHGAC39^{+/0}$ BM recipient mice compared with those that were reconstituted with $V_HJ558^{+/0}$ BM cells (Fig. 5E). Moreover, mature CD45.2$^+$ splenic B cells in $V_HHGAC39^{+/0}$ BM recipients coexpressed endogenous IGH rearrangements at a higher frequency than observed in $V_HJ558^{+/0}$ BM recipients (Fig. 5E, right panels). Additionally, $HGAC39^{+/0}$-expressing B cells exhibited predominantly CD21^{Lo}CD23^{Lo} phenotypes. This was in marked contrast to V_H $J558^{+/0}$-expressing B cells, which exhibited CD21^{Hi}CD23^{Lo} surface phenotypes and tissue localization consistent with the reported MZ B cell differentiation characteristic of this TG (Fig. 5F) (13). The PerC GAC-binding B cell frequency was elevated in V_H HGAC39-chimeric but reduced in V_HJ558 reconstituted mice

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compared with those that received WT only BM (Fig. 5G). CD5 expression was increased on HGAC39-derived GAC-binding B cells relative to V_HJ558-expressing DEX-binding B cells and CD45.1$^+$ WT B cells (Fig. 5H, 5I). We therefore concluded that although V_HHGAC39-expressing B cells were competitively disadvantaged relative to V_HJ558$^{+/0}$ and non-TG B cells during development or maturation, the V_HHGAC39 transgene promoted differentiation of GAC-binding B cells into the B-1 B cell compartment.

Discussion

We show in this study that B cell restricted expression of an IgH TG derived from a GlcNAc-reactive hybridoma in mice promotes surface phenotype and tissue distribution consistent with the innate-like B-1 and MZ B cell development. V_HHGAC39$^{+/0}$ B cells are functionally responsive to BCR ligation, and GAS vaccine immunization drives activation marker upregulation, relocation of splenic GlcNAc-specific cells, and the production of GAC-reactive Abs. However, we observed a breakdown in allelic exclusion in V_HHGAC39$^{+/0}$ B cells in adult mice, indicated by a high frequency of B cells expressing the endogenous IgM allele, suggestive of negative selection in B cell development resulting in secondary gene rearrangement in immature B cells. Indeed, we found that peripheral tolerance checkpoints at transitional B cell stages impeded V_HHGAC39$^{+/0}$ B cell maturation and disadvantaged V_HHGAC39$^{+/0}$ B cells in mixed BM chimeras. These developmental effects are related to HGAC39 specificity because allelic exclusion is maintained in J558 and VH81x TG mice that were generated using similar TG constructs.

B-1 B cells predominantly develop during fetal and neonatal B cell lymphopoiesis from unique yolk sac–derived B cell precursors and have long been known to exhibit enrichment for autoreactive B cell clonotypes (20, 21). Lin28b/Let7 axis-mediated regulation of Arid3a transcription factors in fetal B cells is a master regulator of this long-appreciated ontogenetic switch in B cell development (22, 23). Moreover, that the unique environment in fetal liver promotes B-1 B cell precursors through alternative developmental pathways of B-1 B cell precursors is born from reduced IL-7/STAT5 signaling in the fetal liver environment, which promotes earlier rearrangement of light-chain genes in B cell precursors, thereby bypassing pre-B cell selection checkpoints and enabling their expression of canonical autoreactive V genes that may otherwise pair poorly with surrogate L chain (24). However, Ag-dependent BCR-dependent signaling remains the ultimate regulator of B-1 B cell differentiation (23, 25), and BCR receptors specific for self-antigens can engender B-1 B cell-like phenotypes and functional changes in mature mouse B cells (26). B cells developing early are recruited into Ag-specific compartments to form a durable, self-sustaining population into adulthood (19). Our finding that passively administered GlcNAc-specific mAb attenuated establishment of GlcNAc-reactive B-1 B cells in young V_HHGAC39$^{+/0}$ mice agrees with a requirement for Ag-dependent selection into the B-1 B cell compartment and strongly indicates the existence of a temporal window during which lymphopoiesis is permissive to development of V_HHGAC39-expressing GlcNAc-reactive B cells. Although the rate of accumulation or the turnover of B1b GAC-reactive cells is unknown, the coordinated inflation of B1a B cells frequencies in mice treated with GlcNAc-specific mAb may reflect an increased sensitivity of B1b cells for Ag selection as we have described (9), resulting in the selective loss of these cells.

Self-tolerance establishment in the nascent B cell repertoire occurs through both central and peripheral mechanisms. Central tolerance is composed of at least three distinct mechanisms affecting primarily immature B cells in BM, including clonal deletion, anergy, and receptor editing (27–29). However, we did not find evidence for deletion of V_HHGAC39$^{+/0}$ clonotypes in V_HHGAC39$^{+/0}$ TG mice, nor did we observe significant increases in λ L chain gene expression among endogenous Ig-expressing B cells, suggesting that central tolerance mechanisms did not contribute significantly to breakdown in allelic exclusion in V_HHGAC39$^{+/0}$ B cells. Rather, our data consistently indicated that peripheral tolerance checkpoints represented the developmental stage at which V_HHGAC39 expression was impacted to the greatest extent. Specifically, we observed the greatest increase in V_HHGAC39$^{+/0}$ B cells expressing endogenous IGH gene rearrangements at the T2 B cell stage, during splenic maturation of immature B cells.

It is clear from numerous studies that BCR-mediated signal transduction is a primary determinant of peripheral B cell tolerance. B cells undergo apoptosis following BCR ligation, whereas T2 B cells upregulate survival factors and enter the cell cycle, suggesting that the T1 checkpoint represents a point of BCR-induced negative selection in the periphery (30). Similarly, recent studies indicate the T1 checkpoint is the stage at which cells are selected into the splenic MZ (31).

Although comparisons between BCR TG mice carry caveats due to potential differences in integration site and copy number of V_H transgenes, which can affect overall expression, we have shown that V_HHGAC39$^{+/0}$ mice exhibit an increased rate of endogenous Ig gene rearrangement at the T2 B cell stage relative to V_HJ558$^{+/0}$ TG mice, which express a DEX-specific BCR under the same promoter and enhancer configuration. Moreover, V_HHGAC39$^{+/0}$ B cells were more disadvantaged in competitive BM chimeric mice than were V_HJ558$^{+/0}$, collectively suggesting that this phenomenon is associated with the specificity of the V_HHGAC39 transgene. We propose, on the basis of these published findings and our analyses included in this article, that the breakdown in allelic exclusion occurring in V_HHGAC39$^{+/0}$ B cells is mediated by BCR-dependent engagement of endogenous GlcNAc-modified Ags. In support of this hypothesis, We also observed accumulation of T3 V_HHGAC39$^{+/0}$ B cells with downregulated IgM expression, a developmental subset associated with BCR anergy (32), and is consistent with prior studies in Ig-knock-in mice wherein expression of BCR reactive with neo-self-antigens resulted in negative selection of transgene-expressing B cells (33).

However, sufficient numbers of T2 B cells in V_HHGAC39$^{+/0}$ mice rearrange endogenous Ig alleles and produce endogenous serum Ab, including prototypic NAb specificities, suggesting that V_HHGAC39$^{+/0}$ undergoing secondary gene rearrangements predominated over those entering anergic states. Studies examining the impact of endogenous Ags on B cell selection indicate that both Ag configuration such as membrane-bound versus soluble HEL Ags on MD4 B cell development, plus Ag abundance, significantly influence these processes and confirm that BCR signal strength can lead to qualitatively distinct outcomes in B cell development (34, 35). Somewhat similarly, Ig knock-in mice expressing a hybridoma-derived BCR specific for alpha1-3 galactose (α-Gal) Ags, commonly implicated in human xenograft rejection, fail to exhibit increased frequencies of circulating Gal-specific B cells or serum Abs. However, ablating the α1-3 GalT gene and the ensuing loss of tolerizing glycan-associated α-Gal determinants facilitates αGal-specific Ab production (36). In this light, we postulate that the valency and/or configuration of GlcNAc epitopes in vivo is insufficient to drive clonal deletion in V_HHGAC39$^{+/0}$ and instead promotes secondary gene rearrangement in immature B cells.

Robust proliferation and plasmablast differentiation by HGAC39-expressing B cells upon GAS but not group C *Streptococcus* immunization suggests that microbial Ag can assist in overcoming these negative selection pressures. Although the relative function of pattern

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recognition receptor expression by immature B cells is unknown, TLRs including TLR9 are expressed at transitional B cell stages in the spleen (37) and may significantly alter the signaling milieu of B cell precursors (38). Moreover, we previously reported that GlcNAc-reactive B cells require stimulation from the commensal microbiota for their development and that the content of the intestinal microflora contributes to clonotypic profiles in the GlcNAc-reactive B cell precursors ultimately represented in the B-1 B cell repertoire (9). Collectively, our present and previous findings illustrate a complex interplay of signals driven by endogenous and exogenous Ags in GlcNAc-reactive B cell development.

Disclosures

The authors have no financial conflicts of interest.

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